

Evaluation Metrics Across AI Systems

A practical table guide with banking examples. Use a held-out test set or reviewed task cases; report sample size and define the positive class and pass criteria before comparing systems.

1. Regression

Predict a number, such as next month's loan repayment amount.

Metric	What it tells you	Best	Banking example
MAE	Average absolute difference between actual and predicted values.	Lower	A MAE of Rs 2,000 means predictions miss the true amount by Rs 2,000 on average.
MSE	Average of squared prediction errors; large misses count heavily.	Lower	A few very large loan amount errors raise MSE sharply.
RMSE	Square root of MSE; error in the target's original units.	Lower	RMSE of Rs 3,000 expresses a typical error scale in rupees.
R-squared (R2)	Share of target variation explained relative to predicting its mean.	Higher	R2 = 0.80 means 80% of observed variation is explained on the test set.
Adjusted R2	R2 adjusted for the number of input features.	Higher	Adding a weak customer field may not improve adjusted R2.
MAPE	Average absolute percentage error, relative to actual values.	Lower	MAPE = 10% suggests a typical relative error near 10%; avoid when actuals are zero or near zero.
Bias / mean error	Average signed prediction error; shows systematic over or under prediction.	Near 0	A positive mean(predicted - actual) indicates overprediction.

2. Classification

Predict a category, such as whether a transaction is fraudulent. Treat fraud as the positive class.

Metric	What it tells you	Best	Banking example
Confusion matrix	Counts true positives (TP), true negatives (TN), false positives (FP), false negatives (FN).	Inspect	Shows caught fraud, correctly cleared transactions, false alarms, and missed fraud.
Accuracy	$(TP + TN) / \text{all cases}$.	Higher	90% accuracy can still miss many fraud cases if fraud is rare.
Precision	$TP / (TP + FP)$: how many flagged transactions truly are fraud.	Higher	High precision means fewer genuine transactions are blocked.
Recall / sensitivity	$TP / (TP + FN)$: how many fraud cases are caught.	Higher	High recall means fewer fraudulent transactions are missed.
Specificity	$TN / (TN + FP)$: how many genuine transactions are cleared.	Higher	High specificity means fewer genuine transactions are flagged.
F1 score	Harmonic mean of precision and recall.	Higher	Useful when both fraud misses and false alarms matter.
ROC AUC	How well scores rank positives above negatives across thresholds.	Higher	Compare fraud scoring models before choosing an alert threshold.
PR AUC / average precision	Precision-recall tradeoff across thresholds; informative for rare positives.	Higher	Often useful when fraud is uncommon.
Log loss	Penalizes wrong or overconfident predicted probabilities.	Lower	A 99% fraud prediction for a genuine transaction is heavily penalized.

Example: TP = caught fraud; FN = missed fraud; FP = genuine transaction flagged; TN = genuine transaction cleared.

3. GenAI

Evaluate generated text against a task, rubric, and representative examples.

Metric	What it tells you	Best	Banking example
Task success	Share of outputs that complete the requested task correctly.	Higher	A reply gives the correct steps to dispute a card transaction.
Factual accuracy	Share of checkable claims that are correct against trusted evidence.	Higher	A generated answer quotes the right fee and eligibility rule.
Instruction following	Share of responses that obey stated requirements and format.	Higher	The answer follows a 3-bullet limit and required tone.
Relevance	How directly the answer addresses the user's question.	Higher	An interest-rate question receives an interest-rate answer.
Completeness	Share of required points covered by the response.	Higher	A loan summary includes rate, term, fees, and eligibility.
Safety / policy compliance	Share of outputs that meet defined safety and organizational rules.	Higher	The assistant avoids exposing an account number in a reply.
Format validity	Share of responses that parse and satisfy the requested schema.	Higher	The returned JSON has the required fields and valid types.
Latency and cost	Response time and spend or tokens per completed request.	Lower	Track seconds and rupees per customer support answer.

4. RAG

Assess retrieval and the final answer separately. Compare with trusted source documents.

Metric	What it tells you	Best	Banking example
Retrieval recall@k	Share of relevant documents found among the top k retrieved.	Higher	The correct loan policy is present in the top 5 passages.
Retrieval precision@k	Share of the top k retrieved documents that are relevant.	Higher	Four of five retrieved policy snippets help answer the question.
Context precision	How relevant the retrieved context is, often accounting for ranking.	Higher	Useful policy passages appear before unrelated ones.
Context recall	How much of the information needed for the reference answer is retrieved.	Higher	Retrieval includes all terms needed to answer eligibility and fees.
Faithfulness / groundedness	Share of answer claims supported by retrieved context.	Higher	A statement about a 12-month term appears in the cited policy.
Answer relevance	How directly the answer responds to the user query.	Higher	The answer explains eligibility instead of unrelated loan products.
Answer accuracy	Whether the final answer matches a trusted reference or verified facts.	Higher	The rate and fee match the current approved policy.
Citation accuracy	Share of cited claims supported by the cited passages.	Higher	Each quoted policy sentence points to its supporting document.

5. Agentic AI

Evaluate the end goal plus the agent's decisions, tool use, and safeguards.

Metric	What it tells you	Best	Banking example
Goal / task success	Share of requests where the intended end state is reached.	Higher	The agent finishes a card replacement request correctly.
Tool selection accuracy	Share of steps using the appropriate tool for the task.	Higher	It checks account status before initiating replacement.
Tool call correctness	Share of calls with correct tool, arguments, and allowed scope.	Higher	It uses the right account ID and replacement reason.
Workflow / step correctness	Share of runs that follow required steps and ordering.	Higher	Identity verification occurs before any account change.
Authorization compliance	Share of actions that respect permissions and approval rules.	Higher	The agent requests approval before a high-value transfer.
Recovery rate	Share of recoverable errors resolved without an incorrect final action.	Higher	After a tool timeout, it retries safely and verifies the result.
Human escalation accuracy	Share of cases correctly escalated when human review is required.	Higher	A disputed high-value transfer is routed to an analyst.
Efficiency	Tool calls, elapsed time, and cost per successful task.	Fewer / lower	Compare calls and spend only among equally successful runs.
Unsafe action rate	Share of runs with unauthorized or harmful attempted actions.	Lower	Count any attempted account change without required verification.

How to use this guide

Choose 2-4 quality metrics and 1-2 operational metrics per use case. For GenAI, RAG, and agents, define a scoring rubric and review a sample of outputs manually. An automated judge score is an estimate, not ground truth. Measure both successful outcomes and unsafe failures.

References: scikit-learn metrics documentation (<https://scikit-learn.org/stable/api/sklearn.metrics.html>); Ragas metrics documentation (https://docs.ragas.io/en/latest/concepts/metrics/available_metrics/). Metric definitions for custom task, safety, and workflow rates are practical rubric suggestions.